

Calculus III Final — Integral Meanings & Theory Notes

Focused on the announced final topics: directional derivatives, absolute extrema/Lagrange multipliers, multiple integrals, polar/spherical coordinates, scalar line integrals, Green's theorem, and vector surface integrals (flux).

1. The Big Idea: What Does an Integral Mean?

An integral is an **accumulation**. The differential tells you what tiny object is being accumulated: dx = tiny length on an axis, dA = tiny area, dV = tiny volume, ds = tiny arc length along a curve, and dS = tiny surface area.

Integral	What is accumulated?	Typical meaning
$\int f(x) dx$	values $\times$ tiny lengths	signed area / total accumulation
$\int_D f dA$	values $\times$ tiny areas	volume, mass of lamina, average over a region
$\int_E f dV$	values $\times$ tiny volumes	mass/charge in a 3-D solid; volume if $f=1$
$\int_C f ds$	values $\times$ tiny arc lengths	mass of a wire if f is linear density
$\int_S f dS$	values $\times$ tiny surface areas	mass of a thin surface if f is surface density
$\int_S \mathbf{F} \cdot \mathbf{n} dS$	normal flow $\times$ tiny surface areas	flux through a surface

2. Double and Triple Integrals

Double integral: $\int_D f(x,y) dA$ adds the contribution $f(x,y)dA$ from every tiny patch of a 2-D region D . If $f \geq 0$, it can represent the volume below $z=f(x,y)$ and above D . In particular, $\int_D 1 dA = \text{Area}(D)$.

Triple integral: $\int_E f(x,y,z) dV$ adds contributions from every tiny volume element in a 3-D solid. In particular, $\int_E 1 dV = \text{Volume}(E)$. If f is density, the integral gives total mass.

Fubini: a multiple integral can be evaluated as iterated single integrals when the usual continuity/integrability conditions hold. The inner integral is performed first while the outer variables are treated as constants.

Likely theory questions:

- What does $\int_D f dA$ represent geometrically when $f \geq 0$? — Volume under $z=f$ and above D .
- What does $\int_D 1 dA$ represent? — Area of D .
- What does $\int_E 1 dV$ represent? — Volume of E .
- If density is $\delta(x,y,z)$, what does $\int_E \delta dV$ give? — Total mass.
- What does the order $dz dy dx$ mean? — Integrate z first, then y , then x .
- Why can changing the order of integration help? — The same region may have much simpler bounds in another order.

3. Polar, Cylindrical, and Spherical Coordinates

Polar: $x=r \cos\theta$, $y=r \sin\theta$, $x^2+y^2=r^2$, and $dA = r dr d\theta$. The extra r is the Jacobian: a small polar “rectangle” gets wider as r increases.

Cylindrical: $x=r \cos\theta$, $y=r \sin\theta$, $z=z$, so $dV = r dz dr d\theta$ (order may vary). Use it for cylinders, paraboloids, and rotational symmetry around the z -axis.

Spherical: $x=\rho \sin\phi \cos\theta$, $y=\rho \sin\phi \sin\theta$, $z=\rho \cos\phi$, with $dV = \rho^2 \sin\phi d\rho d\phi d\theta$. Here ρ is distance from the origin, θ is the xy -plane angle, and ϕ is measured downward from the $+z$ -axis.

Memory: circle $\rightarrow$ polar; cylinder/vertical rotational solid $\rightarrow$ cylindrical; sphere/cone centered at origin $\rightarrow$ spherical.

Potential theory traps: forgetting the Jacobian r or $\rho^2 \sin\phi$; confusing θ and ϕ ; using $0 \leq \phi \leq 2\pi$ (normally ϕ ranges 0 to π).

4. Scalar Line Integral — Type 1

For a scalar field f along a curve C , $\int_C f \, ds$ adds f times tiny pieces of *arc length*. If $\mathbf{r}(t)$ parameterizes C , then $ds = |\mathbf{r}'(t)|dt$.

Formula: $\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt$.

Physical meaning: if f is the linear density (kg/m) of a curved wire, then $\int_C f \, ds$ is its total mass. If $f=1$, the integral is simply the length of C .

Key theoretical fact: because ds is positive arc length, reversing the orientation of C does **not** change a type-1 scalar line integral.

Likely theory questions:

- What is ds ? — An infinitesimal arc length; $ds=|\mathbf{r}'(t)|dt$.
- What is $\int_C 1 \, ds$? — The length of C .
- What does $\int_C \rho \, ds$ mean if ρ is linear density? — Mass of a wire.
- Does reversing C change $\int_C f \, ds$? — No.
- Why is $|\mathbf{r}'(t)|$ present? — It converts a parameter increment dt into physical arc length ds .

5. Green's Theorem

Green's theorem converts a **closed line integral around a planar boundary** into a **double integral over the enclosed region**. For positive (counterclockwise) orientation:

$$\oint_C P \, dx + Q \, dy = \iint_D (\partial Q/\partial x - \partial P/\partial y) \, dA.$$

Meaning: the total circulation around the boundary equals the accumulated microscopic rotation (2-D curl) inside the region.

When can you use it? C should be a positively oriented, piecewise-smooth, simple closed curve bounding D , with P and Q having appropriate continuous partial derivatives on an open region containing D .

Orientation: positive = counterclockwise. If C is clockwise, insert a minus sign or reverse the orientation.

Likely theory questions:

- What does "positively oriented" mean in Green's theorem? — Counterclockwise; the region stays on your left as you traverse C .
- Why is Green's theorem useful? — A difficult boundary integral may become an easy area integral, or vice versa.
- Can Green's theorem be used on an open curve? — Not directly; the standard theorem requires a closed boundary.
- What happens if orientation is reversed? — The line integral changes sign.
- What quantity appears inside the double integral? — $\partial Q/\partial x - \partial P/\partial y$, the scalar 2-D curl/circulation density.

6. Vector Surface Integral / Flux — Type 2

For a vector field $\mathbf{F}$ and an oriented surface S , **flux** is

Flux = $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.

The dot product keeps only the component of $\mathbf{F}$ that is **perpendicular (normal)** to the surface. Tangential flow contributes zero flux.

Physical meaning: imagine $\mathbf{F}$ as fluid velocity. Flux measures the net amount/rate of fluid crossing the surface in the chosen normal direction. For a closed surface with outward normal: positive flux means net outward flow; negative means net inward flow; zero means inflow and outflow cancel overall (not necessarily that no fluid crosses).

Orientation matters: replacing $\mathbf{n}$ by $-\mathbf{n}$ changes the sign of the flux.

Graph shortcut: if S is $z=g(x,y)$, then an upward non-unit normal area vector is $\mathbf{n} = -g_x \mathbf{i} - g_y \mathbf{j} + \mathbf{k} \, dA$; for downward orientation use its negative.

Likely theory questions:

- What is flux? — Net normal flow through a surface.
- Why $F \cdot n$? — Dot product extracts the normal component of F .
- What if F is tangent to S everywhere? — Flux is 0.
- What if F points exactly along n ? — Flux density is $|F|$ (maximal positive crossing).
- What does negative flux mean? — Flow is predominantly opposite the chosen normal direction.
- Does changing orientation affect flux? — Yes; it multiplies the integral by -1 .
- For a closed surface, what does “outward orientation” mean? — Normals point away from the enclosed solid.

7. Derivative Theory That Could Be Asked

Directional derivative: $D_u f(P) = \nabla f(P) \cdot u$, where u must be a **unit vector**. It is the instantaneous rate of change of f at P when moving in direction u .

Gradient: ∇f points in the direction of steepest increase. The maximum directional derivative is $|\nabla f|$; the minimum is $-|\nabla f|$ in the opposite direction. A direction perpendicular to ∇f gives directional derivative 0.

Lagrange multipliers: for extrema of f subject to $g=k$, solve $\nabla f = \lambda \nabla g$ together with $g=k$. At a regular constrained extremum, the level curve/surface of f is tangent to the constraint, so their normals (gradients) are parallel.

Absolute extrema on a closed bounded domain: do not only use Lagrange multipliers. Check (1) interior critical points, (2) every boundary piece, including Lagrange candidates on smooth constraint boundaries, and (3) corners/endpoints. Compare all f -values.

Likely theory questions:

- Why must the direction vector be normalized? — Directional derivative is defined per unit distance traveled.
- What direction gives the greatest increase? — $\nabla f/|\nabla f|$.
- What is the greatest rate of increase? — $|\nabla f|$.
- Why does $\nabla f = \lambda \nabla g$ appear in constrained optimization? — At a regular constrained extremum the level set of f is tangent to the constraint, so their normal vectors are parallel.
- Does Lagrange multipliers automatically find every absolute extremum on a filled-in closed region? — No. It handles constraint boundaries; interior critical points and nonsmooth boundary points must also be checked.
- Why is “closed and bounded” important? — For a continuous function, the Extreme Value Theorem guarantees absolute maximum and minimum values exist.

8. Fast Recognition: What Integral Am I Looking At?

Notation / wording	Interpretation / method
$\int_C f \, ds$	Scalar line integral; accumulate scalar along arc length.
$\oint_C P \, dx + Q \, dy$	Vector line integral / circulation; if C is closed planar boundary, consider Green.
$\iint_D f \, dA$	Double integral over area; polar if circular geometry.
$\iiint_E f \, dV$	Triple integral over volume; cylindrical/spherical if symmetry fits.
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	Flux / surface integral type 2; orientation is essential.
“mass of wire”	$\int_C \text{density} \, ds$.
“mass of lamina”	$\iint_D \text{density} \, dA$.
“mass of solid”	$\iiint_E \text{density} \, dV$.
“net flow through surface”	Flux $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$.

9. High-Yield Conceptual Traps

- **ds vs dr:** ds is scalar arc length; dr is a directed displacement vector.
- **Scalar line integral orientation:** $\int f \, ds$ does not change when direction reverses; $\int \mathbf{F} \cdot d\mathbf{r}$ does.
- **Flux is not surface area:** $\iint \mathbf{F} \cdot \mathbf{n} \, dS$ weights each surface patch by the normal component of $\mathbf{F}$.
- **Zero flux $\neq$ zero field:** the field may be tangent to the surface or positive/negative contributions may cancel.
- **Polar/spherical Jacobians are mandatory:** $dA = r \, dr \, d\theta$ and $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$.
- **Green needs a closed boundary:** if a curve is open, you may sometimes add a segment to close it, but must account for that added piece.
- **Absolute extrema:** finding $\nabla f = 0$ only checks interior candidates; the boundary can contain the actual maximum/minimum.

10. Mini Oral/Theory Drill

Try answering each in one or two sentences without calculations.

1. Explain the geometric meaning of a double integral of a nonnegative function.
2. Why does dA become $r \, dr \, d\theta$ in polar coordinates?
3. What is the difference between a scalar line integral and flux?
4. Why does reversing a scalar line integral path not change its value?
5. State Green's theorem and its orientation requirement.
6. Explain physically what positive, negative, and zero flux mean.
7. Why does only the normal component of a vector field contribute to flux?
8. What does the gradient tell you geometrically?
9. Why are gradients parallel in the Lagrange multiplier method?
10. List every category of candidate you must check for absolute extrema on a closed bounded planar region.

Exam memory line: Integrals accumulate over a geometric object. Read the differential first: $dA \rightarrow$ area, $dV \rightarrow$ volume, $ds \rightarrow$ curve length, $dS \rightarrow$ surface area; adding $\mathbf{F} \cdot \mathbf{n}$ turns a surface integral into oriented flux.